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Exercise 1

Quantum phase estimation (QPE) can be used to measure the Hamming weight of a quantum state and prepare Dicke states.

a) Write down the Hamiltonian whose eigenvalues can distinguish the Hamming weight of an n qubit quantum state.

The Hamiltonian whose eigenvalues differs by Hamming weight in n-qubit system can be constructed by the projector onto the space of fixed Hamming weight i. That is,

$$\mathcal{H} = \sum_{i=0}^{2^n-1} e_i P_i$$

We can choose $e_i = i$ where the eigenvalue becomes the Hamming weight i. Note that the given hamiltonian can have ${}_nC_i$ for e_i.

b) For an n qubit state, what is the minimum number of bits needed to measure the hamming weight of the state? Hint: The Hamming weight of the state is in the range $\{0, 1, .., n\}$

Let k denote the minimum number of bits required to measure hamming weight of the state. Then

$$k = [\log_2(n+1)]$$

where $[a]$ denotes the smallest integer larger than a.

Exercise 2

Using qiskit, create a function that takes in a quantum state in vector format as input and returns a quantum circuit that measures the Hamming weight of the state. Hint: see ref [1]. You can use qiskit's initialize module to initialize the states.

Exercise 3

Use the function created in part 2 above to measure the Hamming weight of the following states:

a) $|11\rangle$

b) $|01\rangle$

c) $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

d) $\frac{1}{\sqrt{3}}(|011\rangle + |101\rangle + |110\rangle)$

e) Explain the result in part c) above.

Exercise 4

Explain how the function in part 2 above can be used to prepare a Dicke state.

Exercise 5

Suppose we want to prepare the Dicke state with $n = 100$ and $k = 50$.

What is the probability that the state will be measured?

What is the minimum number of measurements required?

How many gates are required?

References

[1] Soorya Rethinasamy, Margarite L. LaBorde, and Mark M. Wilde. Logarithmic-depth quantum circuits for hamming weight projections, 2024.